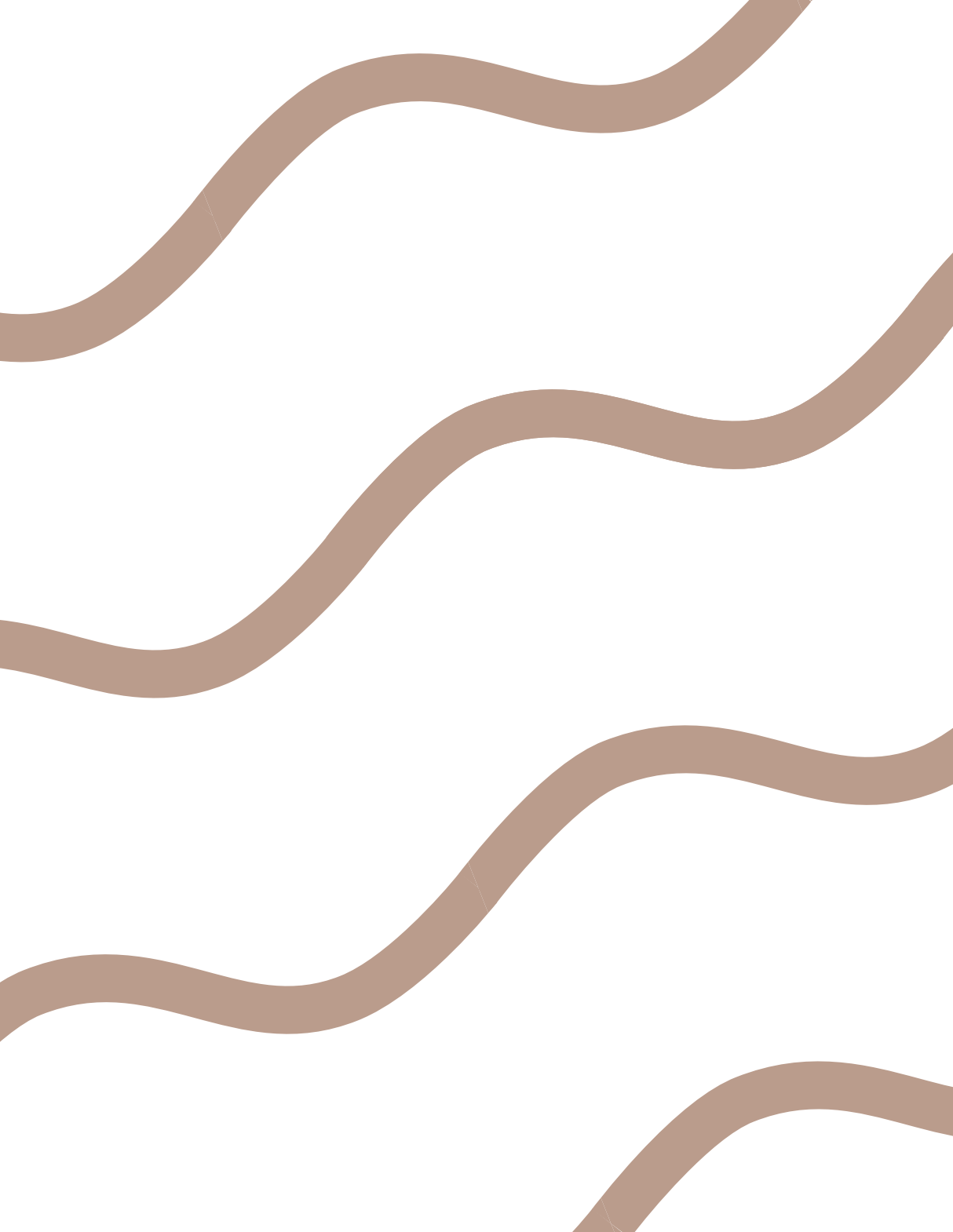




22. (本题满分 12 分)

设二次型 $f(x_1, x_2) = x_1^2 - 4x_1x_2 + 4x_2^2$, $g(x_1, x_2)$ 的二次型矩阵为 $B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

(1) 是否存在可逆矩阵 D , 使 $B = D^T D$? 若存在, 求出矩阵 D ; 若不存在, 请说明理由;

(2) 求 $\max_{x \neq 0} \frac{f(x)}{g(x)}$, 其中 $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

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22.【解】 (1) 由于二次型矩阵 $B = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, $B^T = B$, 其中 $1 > 0$, $\begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} > 0$, 故

B 为正定矩阵. 又 $g(x_1, x_2) = (x_1 - x_2)^2 + x_1^2$, 令 $\begin{cases} x_1 - x_2 = y_1, \\ x_2 = y_2, \end{cases}$ 于是有

$$\begin{cases} x_1 = y_1 + y_2, \\ x_2 = y_2, \end{cases} \text{ 即 } x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = Cy, \text{ 使}$$

$$x^T B x = y^T C^T B C y = y^T y,$$

即

$$C^T B C = E,$$

所以

$$B = (C^{-1})^T C^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}^T \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = D^T D.$$

故存在可逆矩阵 $D = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, 使 $B = D^T D$.

(2) 设 $f(x)$ 的二次型矩阵为 A , 则

$$\frac{f(x)}{g(x)} = \frac{x^T A x}{x^T B x} = \frac{x^T A x}{x^T D^T D x} = \frac{x^T A x}{(Dx)^T Dx} \stackrel{\text{令 } Dx = z}{=} \frac{x = D^{-1}z}{z^T z} \frac{(D^{-1}z)^T A D^{-1}z}{z^T z} = \frac{z^T (D^{-1})^T A D^{-1}z}{z^T z},$$

$$\text{其中 } z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}.$$

$$\text{又 } (D^{-1})^T A D^{-1} = C^T A C = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \stackrel{\text{记}}{=} M.$$

其对应的二次型为 $h(z_1, z_2) = z_1^2 + z_2^2 - 2z_1 z_2$, 则由

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$$|\lambda E - M| = \begin{vmatrix} \lambda - 1 & 1 \\ 1 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 1 = \lambda(\lambda - 2) = 0,$$

$$\text{得 } \lambda_1 = 2, \xi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}; \lambda_2 = 0, \xi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \text{ 令 } Q = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, z = Qr = Q \begin{pmatrix} r_1 \\ r_2 \end{pmatrix},$$

则

$$\begin{aligned} z^T M z &= r^T Q^T M Q r = r^T \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} r = 2r_1^2 \\ &\leq 2 \quad (r_1^2 + r_2^2 = r^T Q^T Q r = r^T r = 2 \quad (z_1^2 + z_2^2)), \end{aligned}$$

$$\text{于是 } z^T M z \leq 2z^T z, \text{ 即 } \frac{z^T M z}{z^T z} \leq 2. \text{ 令 } z_0 = Q \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \text{ 得 } \frac{(1,0) \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}}{(1,0) \begin{pmatrix} 1 \\ 0 \end{pmatrix}} = 2, \text{ 此时 } x_0 =$$

$$D^{-1} z_0 = D^{-1} Q \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ \sqrt{2} \end{pmatrix}, \text{ 于是 } \max_{x \neq 0} \frac{f(x)}{g(x)} = 2.$$

【注】 在 2025 版《张宇线性代数 9 讲》中已经指出, $\frac{x^T A x}{x^T B x}$ 比 $\frac{x^T A x}{x^T x}$ 使用范围更为广泛, 后者已经出现过, 前者尚未出. 在真题中, 值得注意.